

COURSE PROJECT REPORT

Course Title: [Real-Time Systems / RTOS]

Smart Controller System using RTOS

Group Members

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Project Requirements (Reference)

Objective: Develop an RTOS-based Smart Climate Control System that monitors temperature and humidity, and controls a heater, cooler, and humidifier (represented by LEDs) to maintain target conditions. The project is a course demo and need not reflect real-world physical operation.

Hardware: YoluUNO (ESP32-S3) microcontroller; DHT20 temperature/humidity sensor on I2C; three two-color LED actuators representing Heater (D3/D4), Cooler (D5/D6), and Humidifier (D7/D8); onboard status LED on GPIO48.

RTOS: FreeRTOS (or any RTOS suitable for the microcontroller), using the task scheduler.

Required behaviors: Serial print of temperature/humidity every 5 seconds; onboard LED blinks every 1 second; Heater LED shows GREEN/ORANGE/RED according to 3 defined temperature thresholds; Cooler activates for a fixed duration (e.g., 5 s) when temperature exceeds a threshold, then re-checks; Humidifier runs a GREEN (5 s) → YELLOW (3 s) → RED (2 s) sequence when humidity falls below a threshold, then re-checks humidity to decide whether to repeat.

Required tasks: Blinky Task, Read Temperature Task, Cooler Task, Heater Task, Humidifier Task.

Deliverable: GitHub repository link and result snapshots.

1. Introduction

2. System Design using State Machine

3. System Implementation

4. Validation Results in Simulation

5. Conclusion and Perspective
